



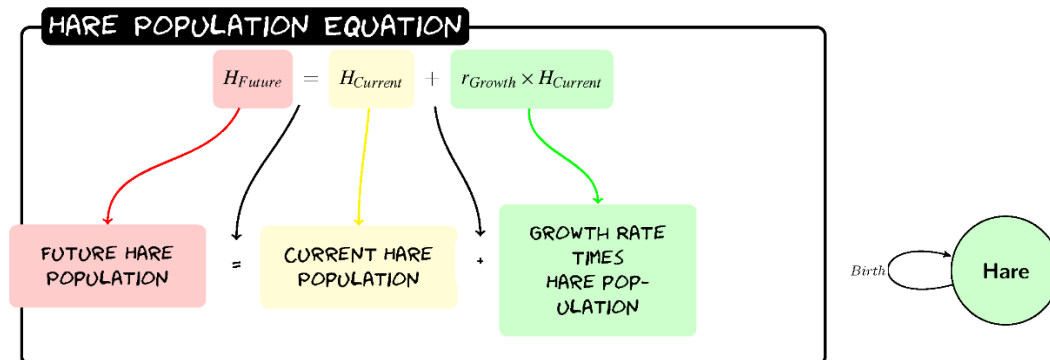
# The Maths of Predators and Preys

## Don't get Eaten by the Wolf

### Hares



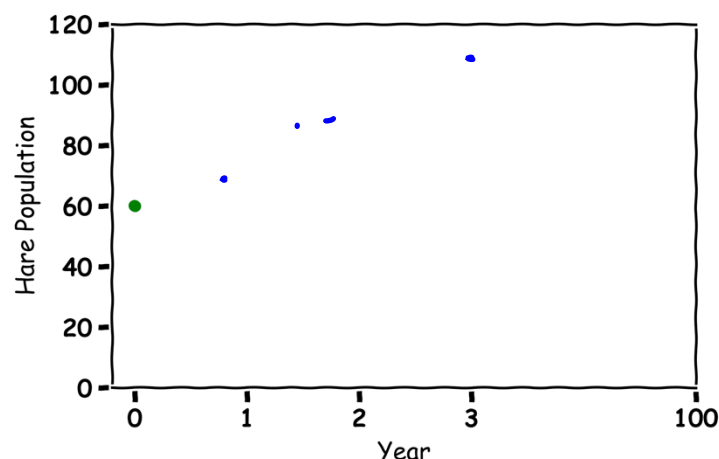
No predators



Given a population of 60 Hares and a growth rate of 0.2 what would the population be after three years?

| Year | Hare Population      |
|------|----------------------|
| 0    | 60                   |
| 1    | $60 + 0.2(60) = 72$  |
| 2    | $72 + 0.2(72) = 86.$ |
| 3    | $86 + 0.2(86) = 103$ |

Sketch the Graph.



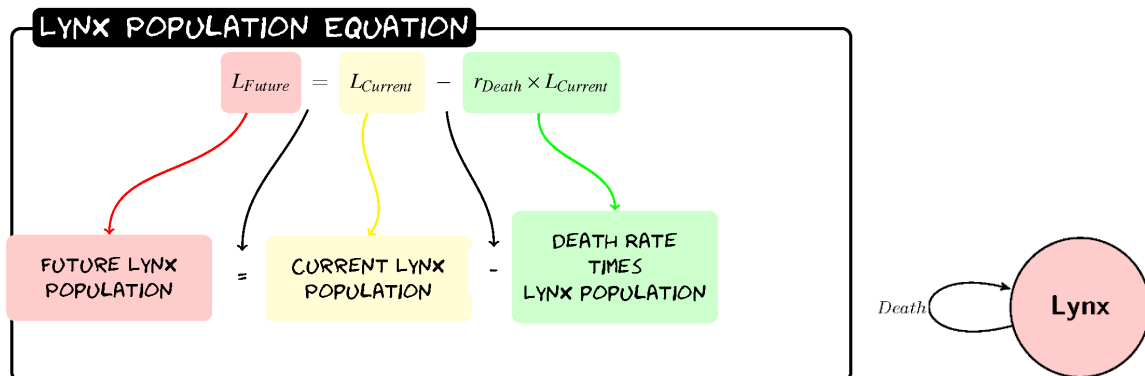
Guess what would happen in 100 years?

over 4 billion



# Lynx

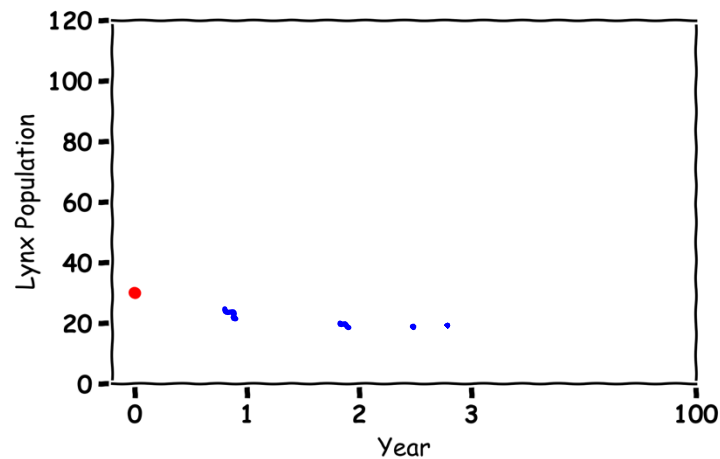
No Prey



Given a population of 30 Lynxes and a death rate of 0.1 what would the population be after three years?

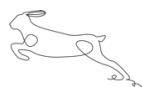
| Year Lynx Population |                                  |
|----------------------|----------------------------------|
| 0                    | 30                               |
| 1                    | $30 - 0.1 \times 30 = 27$        |
| 2                    | $27 - 0.1 \times 27 = 24.3$      |
| 3                    | $24.3 - 0.1 \times 24.3 = 21.87$ |

Sketch the Graph.



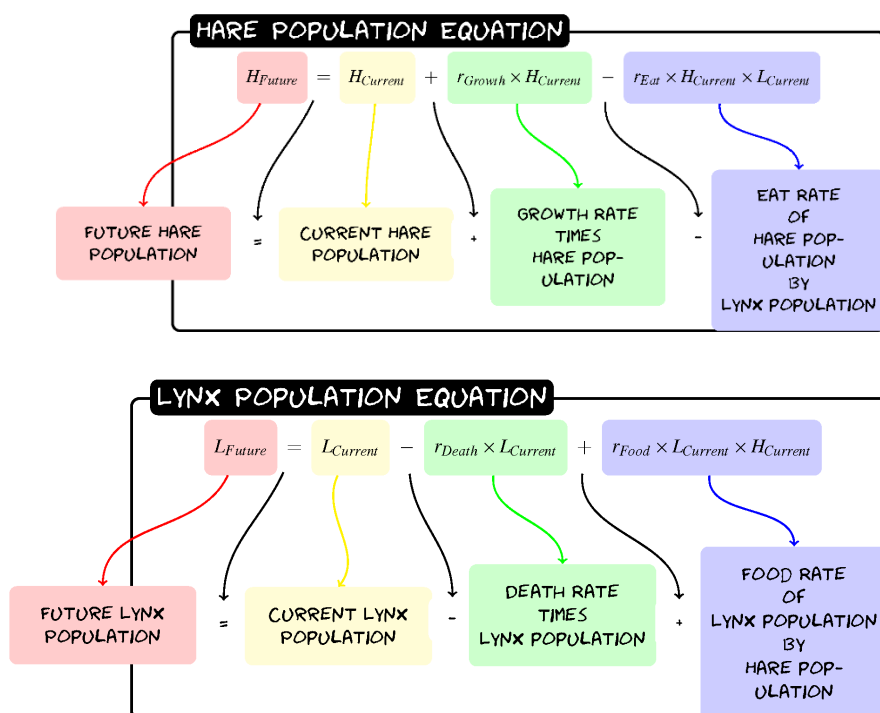
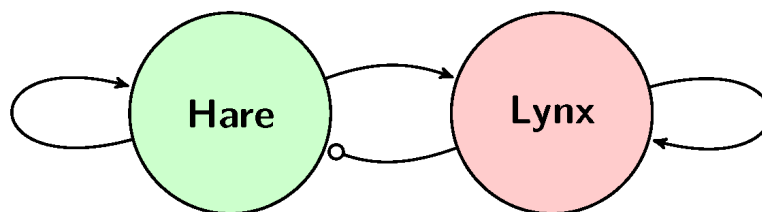
Guess what would happen in 100 years?





## Hares and Lynxes

Label the graph:



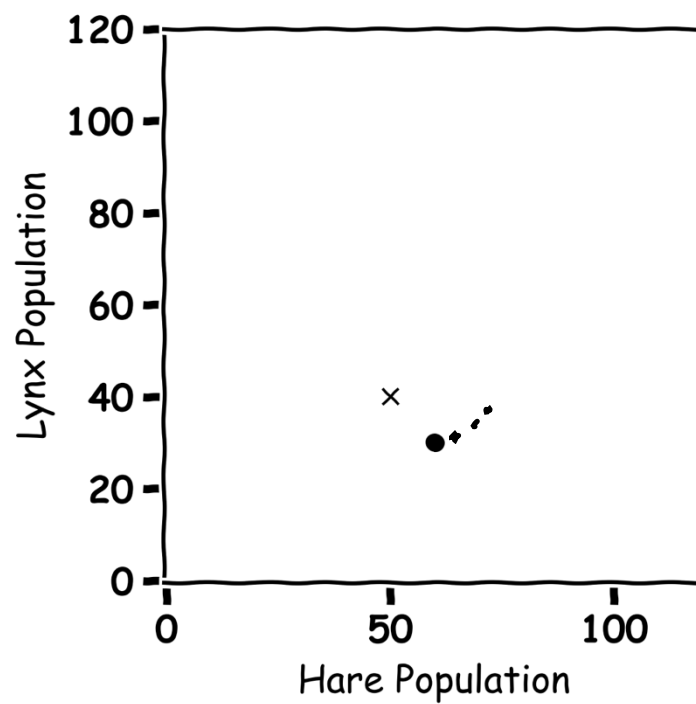
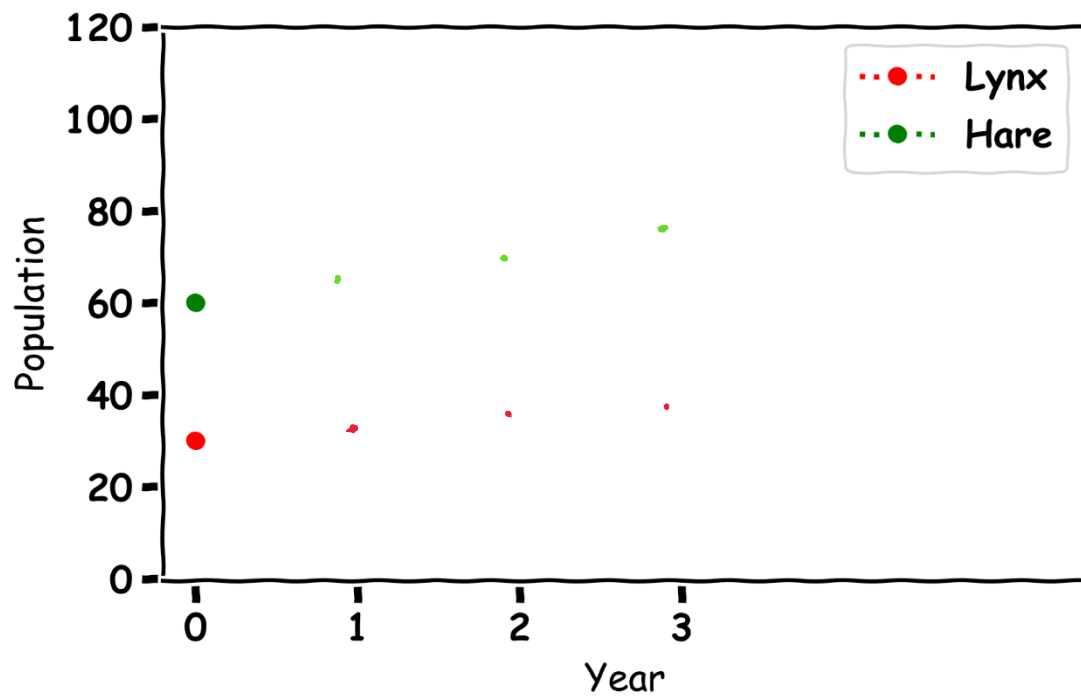
Given a population of 60 Hares and 30 Lynxes with a birth rate of .2, a death rate of 0.1, a eat rate of 0.005 and a food rate of 0.002 what would the populations be after three years?

| Hare   | Hare  | Lynx  | Value |
|--------|-------|-------|-------|
| Growth | 0.2   | Death | 0.1   |
| Eat    | 0.005 | Food  | 0.002 |

| Year | Hare Population                     | Lynx Population                     |
|------|-------------------------------------|-------------------------------------|
| 0    | 60                                  | 30                                  |
| 1    | $60 + 0.2(60) - 0.005(60)(30) = 63$ | $30 - 0.1(30) + 0.002(30)(60) = 31$ |
| 2    | 66                                  | 31                                  |
| 3    | 69                                  | 32                                  |



Sketch the graph:





# Don't get Eaten by the Wolf

Label the graph

